

The Bisection Method

Root-Finding for Nonlinear Equations, with an Application to Orbital Mechanics

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The Problem: Nonlinear Equations

We want to find x^* such that

$$f(x) = 0$$

where f has no closed-form inverse – e.g. $f(x) = x^3 - x - 2$, or $f(x) = e^x - 3x$. Unlike a linear system, there is no finite algebraic procedure that gives the exact answer.

Why an iterative method?

We build a sequence $x_0, x_1, x_2, \dots$ that converges to the true root x^* . The bisection method is the simplest and most robust of these methods: it does not require f to be differentiable, only continuous – unlike Newton-Raphson, which needs f' .

The Intermediate Value Theorem

Intermediate Value Theorem

If f is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$ (opposite signs), then there exists $c \in (a, b)$ such that $f(c) = 0$.

Intuition: a continuous curve cannot jump from negative to positive without crossing zero somewhere in between.

The key simplification

Instead of solving $f(x) = 0$ algebraically, we turn the problem into one of *localization*: shrink an interval that is guaranteed to contain the root, using only the *sign* of f at the endpoints – nothing about the shape of f in between is needed.

Necessary Condition: the Sign Change

The theorem requires $f(a) \cdot f(b) < 0$. This is not optional:

- If $f(a)$ and $f(b)$ have the **same** sign, there is no guarantee a root exists in (a, b) .
- There could be zero roots, or an even number of them that cancel out in terms of net sign change.

Key idea

Before applying bisection, we must always verify this sign change – typically by evaluating f at several points or plotting it.

Bisecting the Interval

At each iteration:

$$c = \frac{a + b}{2}$$

$$\begin{cases} f(a) \cdot f(c) < 0 & \Rightarrow \text{root in } (a, c), b \leftarrow c \\ f(c) \cdot f(b) < 0 & \Rightarrow \text{root in } (c, b), a \leftarrow c \\ f(c) = 0 & \Rightarrow \text{exact root found} \end{cases}$$

Why this is valid at every step

The new interval still satisfies the IVT hypothesis (opposite signs at its endpoints), so the theorem guarantees the root is still inside. “The interval contains a root” is an *invariant* preserved by construction, not an assumption.

Stopping Criterion

Iterate until

$$|b - a| < \varepsilon \quad \text{or} \quad |f(c)| < \delta$$

where ε is the desired tolerance in x , and δ the desired tolerance in f .

Preview

Later in this talk we solve a real physics problem this way: **Kepler's equation** for the eccentric anomaly of an orbit – a root-finding problem with no closed-form solution, dating back to 1609.

Convergence Rate

The interval width shrinks by exactly half every iteration:

$$b_n - a_n = \frac{b_0 - a_0}{2^n}$$

Why exactly half?

The midpoint, by definition, splits the interval into two equal parts – the new interval cannot be anything other than half of the previous one. This is why bisection has **linear convergence** (order 1): the error shrinks by a constant factor each step, regardless of how f behaves.

Iterations needed for tolerance ε :

$$n > \log_2 \left(\frac{b_0 - a_0}{\varepsilon} \right)$$

This bound is known *before* running the algorithm – a guarantee Newton-Raphson does not offer (it may converge faster, but can also diverge for a bad initial guess).

Simple Example: Setup

Find the root of

$$f(x) = x^3 - x - 2$$

Check $a = 1$, $b = 2$:

$$f(1) = -2, \quad f(2) = 4 \quad \Rightarrow \quad f(1) \cdot f(2) < 0$$

So the IVT guarantees a root in $(1, 2)$.

Simple Example: Iterations

n	a	b	c	$f(c)$
0	1.0000	2.0000	–	–
1	1.0000	1.5000	1.5000	–0.1250
2	1.5000	1.7500	1.7500	1.6094
3	1.5000	1.6250	1.6250	0.6660
4	1.5000	1.5625	1.5625	0.2521

Key idea

The interval width is cut exactly in half every row, confirming the formula from the previous slide. The true root is $x^* \approx 1.5214$.

Kepler's Equation

For a body in an elliptical orbit, Kepler's second law (equal areas in equal times) leads to:

$$M = E - e \sin E$$

where M is the **mean anomaly** (proportional to time), e is the **orbital eccentricity**, and E is the **eccentric anomaly** we want to find.

Why bisection?

This equation has **no closed-form solution** for E in terms of elementary functions – E appears both linearly and inside a sine. Kepler posed this exact problem in 1609; it is one of the oldest root-finding problems in physics.

We solve $f(E) = E - e \sin E - M = 0$.

Kepler's Equation: Numerical Setup

Take an orbit with eccentricity $e = 0.7$, at the instant where the mean anomaly is $M = 1.0$ rad. Since $E \in [0, 2\pi)$ for any $e < 1$, we bracket:

$$a = 0, \quad b = \pi$$

$$f(0) = -M = -1.0 \quad f(\pi) = \pi - M \approx 2.1416$$

Opposite signs $\Rightarrow$ IVT guarantees a root in $(0, \pi)$.

Kepler's Equation: Result

Running bisection to tolerance 10^{-10} takes 35 iterations and gives:

Result

$$E \approx 1.69463891 \text{ rad}, \quad f(E) \approx -1.4 \times 10^{-11}$$

From E , the **true anomaly** θ (the actual angular position of the body, seen from the focus) follows from a standard identity:

$$\theta = 2 \arctan \left(\sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2} \right) \approx 139.29^\circ$$

Bisection: Function Signature and Validation

```
1 fa, fb = f(a), f(b)
2
3 # The Intermediate Value Theorem requires a genuine sign change.
4 # Without this, there is no guarantee a root exists in [a, b].
5 if fa * fb > 0:
6     raise ValueError("f(a) and f(b) must have opposite signs "
7                       f"(got f(a)={fa:.4g}, f(b)={fb:.4g}).")
```

Key idea

The sign-change check is not optional: without it, the IVT gives no guarantee that a root exists in $[a, b]$.

Bisection: Main Loop

```
1 history = []
2 for n in range(1, max_iter + 1):
3     # The midpoint is the unique point that splits the interval
4     # into two equal halves -- this is what guarantees the width
5     # is exactly cut in half at every iteration.
6     c = (a + b) / 2.0
7     fc = f(c)
8
9     history.append({"n": n, "a": a, "b": b, "c": c, "fc": fc})
10    if verbose:
11        print(f"n={n:2d}  a={a:.6f}  b={b:.6f}  "
12              f"fc={c:.6f}  f(c)={fc:.6e}  width={b-a:.2e}")
```

Bisection: Updating the Bracket

```
1  if fc == 0.0 or (b - a) / 2.0 < tol:
2      return c, history
3
4  # Keep whichever half still brackets the root: the invariant
5  # "this interval contains a root" is preserved by this choice,
6  # since the new endpoints still have opposite-sign f-values.
7  if fa * fc < 0:
8      b, fb = c, fc
9  else:
10     a, fa = c, fc
```

Key idea

This is the invariant from Section 4: whichever half we keep still has opposite-sign endpoints.

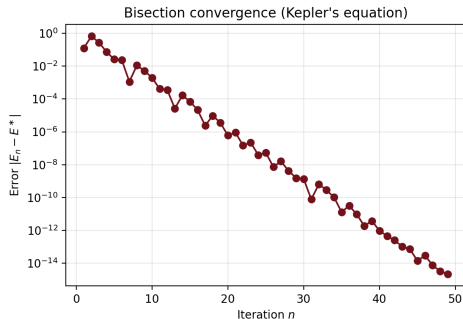
Physics Application: Kepler Residual Function

```
1 def kepler_residual(E, M, e):
2     """f(E) = E - e sin(E) - M; its root is the eccentric anomaly."""
3     return E - e * np.sin(E) - M

1     e = 0.7          # orbital eccentricity (dimensionless, 0 <= e < 1)
2     M = 1.0          # mean anomaly (radians)

3
4     f = lambda E: kepler_residual(E, M, e)
5
6     # For any e < 1, the root always lies in [0, 2*pi]; we bracket
7     # more tightly around M since E is close to M for small-to-moderate e.
8     a, b = 0.0, np.pi
```

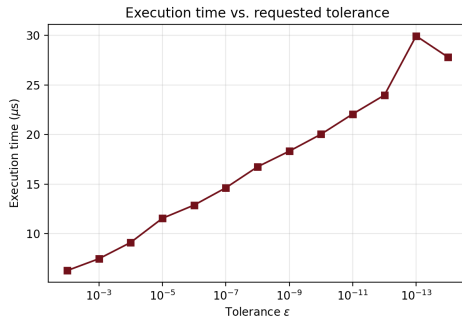
Convergence Plot: Error vs. Iteration



Interpretation

The straight line on a log scale confirms **linear convergence**: the error is reduced by a constant factor (one half) every iteration, exactly as predicted by $b_n - a_n = (b_0 - a_0)/2^n$.

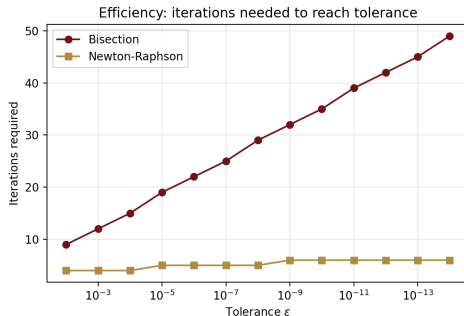
Execution Time vs. Tolerance



Interpretation

Time grows linearly with the number of iterations, which in turn grows as $\log_2(1/\epsilon)$ – tightening the tolerance by orders of magnitude costs only a handful of extra iterations, each a single function evaluation.

Efficiency: Bisection vs. Newton-Raphson



Interpretation

Newton-Raphson (quadratic convergence) needs far fewer iterations than bisection (linear convergence) for the same tolerance. The trade-off: Newton requires f' and a good initial guess; bisection only requires a valid bracket and never diverges.

Summary

- ➊ Bisection rests on the Intermediate Value Theorem: it needs only continuity and a sign change, not differentiability.
- ➋ Each iteration preserves the invariant that the current interval brackets a root – this is what guarantees convergence.
- ➌ Convergence is linear: the error halves every iteration, and the number of iterations needed is known in advance.
- ➍ Applied to Kepler's equation, bisection solves a problem with no closed-form solution, central to orbital mechanics since 1609.
- ➎ Compared to Newton-Raphson, bisection trades speed for robustness: more iterations, but guaranteed convergence.

Thank you

Questions?